

Assignment 4 Answer Key

1. 00000111
2. -7, -7 and -7 respectively
3. 3-input AND gate
4. $F = (A + B + C'). (A + B' + C'). (A' + B + C'). (A' + B' + C).$ $(A' + B' + C')$
5. $B(C+A)$
6. $C = 1$, A & B can be anything
7. $H = 1333.3 \text{ AT/m}$, $B = 1.67 \text{ mT}$, $\phi = 0.84 \mu\text{Wb}$
8. $1.49 \times 10^3 \text{ AT}$
9. $S = 3.18 \times 10^6 \text{ AT/Wb}$, $I = 1.27 \text{ A}$
10. 0.1675 T
11. 0.04 mWb
12. $S = 1.99 \times 10^6 \text{ AT/Wb}$, $B = 3.75 \text{ T}$, $H = 14920.77 \text{ AT/m}$, $N \approx 995$
13. 20 J
14. (a) $1.59 \times 10^6 \text{ AT/Wb}$ (b) 6.37 A
15. $10 \pm 1\%$
16. $\frac{3s+2}{s-1}$
17. $\frac{s+1}{s+2}$